GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

# Followup 26-03

Agricultural crush injury with pelvic hemorrhage and an occult cervical injury

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|---|---|
| **Patient** | 63-year-old male, farm worker, struck by falling hay bales at work. Prior anterior cervical fusion with acquired fusion of the cervical spine, chronic paresthesias of both hands and feet at baseline, HLA-B27 positive, chronic anemia and thrombocytopenia. |
| **Outcome** | Survived. Traumatic subarachnoid hemorrhage, frontal calvarial and nasal fractures, highly comminuted right acetabular and comminuted left pubic fractures, extraperitoneal pelvic hemorrhage, right distal radius and hamate fractures, and a C7-T1 hyperextension injury with anterior longitudinal ligament disruption and a 4 mm epidural spinal hematoma. |
| **Disposition** | Non-operative throughout apart from laceration repair. To inpatient rehabilitation on hospital day 8, neurologically intact, in a cervical orthosis for eight to twelve weeks. |

## The call

This one ends well. He took three quarters of a ton of hay, arrived at a trauma center with a bleeding pelvis and a broken neck nobody had found yet, and walked into rehab eight days later.

Interfacility transfer of a level 2 trauma patient to a trauma center, called for monitored vital signs in a patient who could not ambulate or tolerate a wheelchair. On contact he was alert and oriented times four, airway patent, skin warm and dry, 2 L by nasal cannula, pain 6 out of 10, cervical collar in place. The sending nurse reported vitals had been stable but the pressure was low, and that he had received hydromorphone and droperidol about two hours prior. He was moved to the stretcher by sheet with cervical spine maintained.

| From contact | BP | MAP | Pulse | Pulse pressure |
|---|---|---|---|---|
| +10 min | 107/69 | 82 | 70 | 38 |
| +30 min | 98/53 | 68 | 67 | 45 |
| +50 min | 95/59 | 71 | 76 | 36 |
| +70 min | 90/58 | 69 | 71 | 32 |
| +80 min | 99/65 | 76 | 82 | 34 |

Eight sets were taken at ten minute intervals; five are shown. SpO2 ran 96 to 98%. He slept most of the transport and answered appropriately when woken.

Two sentences in that narrative are the reason this case is worth teaching. The crew documented that the low pressure could be from the hydromorphone and the droperidol, and that the pulse pressure remained greater than 30. The crew was watching the blood pressure, reasoning about where it came from, and measuring it against a number they had picked to reassure themselves with. Most of what follows is about that number.

## Primary impression

Stable interfacility trauma transfer with medication-related hypotension.

Defensible based on the information available to the crew, and the pharmacology is right: hydromorphone lowers blood pressure through histamine release and reduced sympathetic tone, droperidol through alpha-1 blockade, and two hours is inside the window for both.

What the trauma center found was extraperitoneal pelvic hemorrhage, larger than on the scan at the sending facility and tracking into the retroperitoneum, with no active extravasation. That space sits behind the peritoneum where a FAST exam cannot see it, and it accepts several liters before pressure rises enough to tamponade. The blood comes from a venous plexus and from the fracture surfaces themselves, which is why the scan found nothing: **no active extravasation means no contrast left a vessel fast enough to pool during the study, and that threshold is arterial.**

The impression was incomplete in the way trauma is always incomplete: the benign explanation was the only one written down. In a trauma patient with a known pelvic fracture, hemorrhage is the first explanation for a soft pressure and the drugs are the second. Write both.

## Hospital course

**Day 0.** Level 2 activation. Neurologically intact on gross motor testing, reporting new numbness and tingling in both hands and feet on a background of chronic paresthesias. Examination found midline tenderness at the low cervical and high thoracic spine, right hip tenderness with a stable pelvis, and facial lacerations, which were repaired. Outside imaging was repeated, and the comparison between the two studies is what showed the hemorrhage was still growing. New four-extremity paresthesias, the prior fusion, a known intraspinal synovial cyst, and midline low cervical tenderness together took the team to MRI. Chemical VTE prophylaxis was held for the intracranial bleed.

**Days 1 through 7.** MRI confirmed a C7-T1 hyperextension injury with anterior longitudinal ligament disruption and a 4 mm epidural spinal hematoma, managed conservatively on stable alignment and an intact examination, with a Miami J orthosis for eight to twelve weeks. The acetabular, pubic, and radius fractures were managed non-operatively with toe-touch weight bearing right lower extremity and non-weight bearing right upper extremity. Hemoglobin drifted from 12.7 to a nadir of 9.3, not transfused.

**Day 8.** To inpatient rehabilitation.

**Final diagnoses:** polytrauma from agricultural crush injury, as listed above.

## Education points

### The spine that could not bend

The **anterior longitudinal ligament** runs down the front of the vertebral bodies and resists extension: when you tip your head back, that ligament is what stops you.

This patient had an anterior cervical fusion from an injury in adolescence, acquired fusion of the cervical spine, and HLA-B27 positivity, the genetic marker associated with ankylosing spondylitis and the related conditions that fuse a spine on their own. The result is a long segment that used to be a chain of small hinges and is now one rigid bar.

A normal spine absorbs force by spreading it across dozens of joints. A fused spine cannot spread anything. It behaves like a long bone, and a long bone under load concentrates its stress at one point: wherever the rigid segment ends and mobile spine begins. **The fusion does not prevent the injury, it relocates it, and it relocates it somewhere predictable.** His rigid segment ended in the lower cervical spine and the first mobile level below was C7-T1. Our narrative is the only document in this chart recording that he was driven face first onto concrete. His head went into extension, the whole force arrived at one level, and the ligament tore.

Three consequences follow. **Low energy causes high grade injury**, so a ground level fall in an ankylosed spine can produce an unstable three-column fracture. **The pattern is hyperextension and it is unstable**, because the front tether is gone and alignment that looks fine lying flat can fail when position changes. **And they get missed**, with delayed diagnosis running fifteen to forty percent in published series. This one was found on MRI, ordered because a trauma attending put hands on a neck and found midline tenderness in a man who was neurologically intact and talking.

> **Further reading on the ankylosed spine.** This is uncommon in EMS education and it takes some more studying to understand well. The Spine Section of the German Society for Orthopaedics and Trauma published recommendations on spine fractures in ankylosing diseases in Global Spine Journal, free to read in full: [https://pmc.ncbi.nlm.nih.gov/articles/PMC6130102](https://pmc.ncbi.nlm.nih.gov/articles/PMC6130102)

### Blood pressure, and the two numbers that are not the answer

The crew charted a pulse pressure greater than 30. That is a good instinct aimed at the wrong yardstick, and I want to take it apart carefully, because the instinct is one worth keeping.

**Systolic** is the peak pressure during ejection, **diastolic** is the trough between beats, and **pulse pressure** is the difference, which tracks stroke volume. Watching pulse pressure narrow is good practice: as volume is lost the peak comes down while the sympathetic response clamps peripheral vessels and drives the diastolic up, so the two ends converge and the pulse pressure narrows before the systolic collapses.

But narrowing is the signal, and our narrative used a threshold instead. Pulse pressure greater than 30 is not a validated cutoff, and **a difference tells you nothing about where the two numbers sit.** Watch it fail on his own vitals. At +30 minutes he was 98/53: pulse pressure 45, clear of the threshold, MAP 68. At +70 minutes he was 90/58: pulse pressure 32, narrower, MAP 69. The pulse pressure moved thirteen points and the MAP moved one. Sharpen it further: 80/50 gives a pulse pressure of 30 with a MAP of 60, while 90/40 gives a pulse pressure of 50 with a MAP of 57. **The wider pulse pressure has the worse perfusion pressure.**

**Mean arterial pressure is the number the organ sees.** The heart spends a third of each cycle in systole and two thirds in diastole, so the average sits much closer to the diastolic, which is where diastolic plus one third of the pulse pressure comes from. That estimate drifts as heart rate climbs, so use the monitor's computed MAP. His sat between 68 and 71 for a solid hour, above the floor of 65 with nothing underneath it, in a man bleeding into his pelvis.

Now read the pulse column: nothing crossed 90. Age blunts the baroreceptor response, opioids reduce sympathetic outflow, and retroperitoneal blood can drive a vagal bradycardia. The cardiac accelerator fibers also leave the cord at T1 through T4, so an injury at or above that level strips sympathetic supply to the heart while vagal tone continues through a nerve that never enters the cord. He had a C7-T1 injury with an epidural hematoma at that level. I am not telling you that is what happened. I am telling you the anatomy, because a high cord injury and hemorrhagic shock look identical from outside.

### Prehospital considerations

**Document a motor and sensory baseline on every spinal patient.** This man had numb hands and feet before the hay bale existed, so nobody could say how much of the new numbness was new. The value of your exam is the comparison it makes possible.

**Assume the interfacility trauma patient is still a trauma patient.** I know a transfer does not feel like a trauma call. Somebody else did the assessment, the imaging is done, and the patient is packaged and going somewhere better. But imaging is a snapshot, the pathology kept running after the scanner stopped, and this man's hemorrhage was larger on arrival than at the sending facility. **A transfer is an unmonitored interval that you are the monitor for.**

## The interventions

#### PREHOSPITAL INTERVENTIONS

**Spinal motion restriction and the sheet move.** A torn anterior longitudinal ligament with a fused segment above it makes a patient unstable in extension specifically, and his alignment held on imaging because he was flat. **A cervical collar does not stabilize a spine.** It limits gross range of motion and reminds everyone touching him. The stabilization is in the handling: coordinated movement, maintained alignment, no independent motion of head relative to torso. The sheet move is the intervention and the collar is the label on it.

**Monitoring.** Eight blood pressures with computed MAPs over eighty minutes at ten minute intervals, pulse oximetry interleaved. That is a properly monitored trauma transfer and it is the entire reason the truck was called. Ten minutes is the right cadence for a patient whose only warning is going to be a trend, and eight sets is what turns a number into a direction. Everything taught in this document about his blood pressure exists because somebody recorded a MAP next to every cuff reading.

## The labs, and what they mean

*Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.*

| Lab | Value | Reference range |
|---|---|---|
| pH, venous | 7.40 | 7.32 to 7.42 |
| Oxygen saturation, venous (%) | 55 | 65 to 85 |
| Bicarbonate (mmol/L) | 23 | 22 to 32 |
| Base excess (mmol/L) | -2.0 | -2.0 to 2.0 |
| Lactate (mmol/L) | 1.5 | Less than or equal to 2.0 |
| Neutrophils, absolute (x10^9/L) | 11.35 | 1.80 to 7.80 |
| Lymphocytes, absolute (x10^9/L) | 0.32 | 1.00 to 4.00 |

| Lab | Value | Reference range |
|---|---|---|
| Hemoglobin (g/dL) | 12.7 | 14.0 to 18.0 |
| Calcium, ionized (mmol/L) | 1.07 | 1.12 to 1.40 |

**Lactate is normal, pH is normal, bicarbonate is normal, and this patient is in compensated shock.** The evidence is in the line almost nobody reads. Arterial blood leaves the heart nearly fully saturated. Tissue extracts what it needs and the rest returns through the veins, so **venous oxygen saturation is the leftovers.** At rest tissue takes a quarter to a third of what arrives, which is why the normal range sits at 65 to 85% rather than near 100. When delivery falls, tissue does not reduce consumption; it pulls a larger fraction from every unit of blood and the leftovers get smaller. His was 55%: extraction up, delivery down, and he was bleeding. **Extraction rises first and lactate rises last**, because lactate only appears once compensation fails and cells switch to anaerobic metabolism.

**The ionized calcium is the finding to take from this panel.** About 40% of blood calcium is bound to albumin and inert; the ionized fraction is the only part that works, and it is required for the clotting cascade, where it is designated factor IV, for myocardial contractility, and for vascular tone. His was 1.07, and **he had received no blood products.** Most people learn hypocalcemia in trauma as a citrate problem from stored blood. This was intrinsic to the injury, and it is associated with greater transfusion requirement, coagulopathy, and mortality: worse clot formation, weaker contraction, and lower vascular tone all make hemorrhage worse, and hemorrhage makes the calcium worse.

## Learning points

1. **Mean arterial pressure is the number the organ sees.** Systolic is a peak the tissue feels for a third of the cycle, and pulse pressure is a difference that says nothing about where the two numbers sit. This patient went from a pulse pressure of 45 with a MAP of 68 to one of 32 with a MAP of 69. Watch pulse pressure narrow, but trend the MAP.
2. **A normal heart rate does not exclude hemorrhage.** He never crossed 90 while bleeding into his retroperitoneum. Age blunts the baroreceptor response, opioids reduce sympathetic outflow, beta blockade caps it, retroperitoneal blood can drive a vagal bradycardia, and a cord injury at or above T4 removes cardiac accelerator supply.
3. **Compensation is invisible in the numbers you can get, and it fails all at once.** His lactate, pH, and bicarbonate were normal while his venous oxygen saturation sat at 55%, because tissue was pulling extra oxygen from every unit of blood to cover a deficit. Extraction rises first, lactate last, and hemoglobin does not fall until dilution catches up.
4. **A fused or ankylosed spine fails at the hinge, and low energy is enough.** A rigid segment cannot spread force, so it behaves like a long bone and delivers everything to the first mobile level below it. Find a stiff spine in the history and lower your threshold for precautions.

*Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at [https://bcmca.github.io/followup/](https://bcmca.github.io/followup/)*